StifFlow

Revised core mathematical formulation

Calibrated local stiffness, a shared prior, and conditional flow matching

Scope. StifFlow uses quantitatively defined local energetic curvature to shape conditional probability paths for molecular conformer generation. A common graph-dependent source remains sampleable without a target conformer; ordinary conditional flow matching (CFM) learns the marginal velocity. This note retains the source, precision interpolation, and CFM backbone of the supplied v1 formulation. It makes the joint target law and rigid-mode treatment explicit, adopts an endpoint-compatible mean schedule, and derives both the local calibration limit and its mathematical costs. These are proposed design choices and algebraic results, not evidence of molecular performance. CFM and Hessian-informed conditional flows are established ideas [1, 2]; the particular construction below is specified and derived here, not attributed to those papers.

1 Coordinates and the physical quantity

For a labeled molecular graph G with $N \geq 2$ atoms, including any specified stereochemistry, use centered Cartesian coordinates

$$\mathcal{X}_G = \left\{ x \in \mathbb{R}^{3N} : \sum_{a=1}^{N} x_a = 0 \right\}, \qquad d = 3N - 3. \tag{1}$$

All identities, inverses, determinants, densities, and derivatives below are restricted to this linear space and represented in an orthonormal basis. Translation is removed; rotation and internal motion remain. This is not a nonlinear quotient by rigid motions. Let $\mathcal{R}$ denote the restriction of $I_N \otimes R$ to $\mathcal{X}_G$, for $R \in \mathrm{SO}(3)$.

Choose an explicit, sufficiently smooth, rigid-motion-invariant reference potential $E_{\mathrm{ref},G}$ and state its energy and length units. At a stationary reference r_i that is stable against vibrational perturbations,

$$H_i = \nabla^2 E_{\mathrm{ref},G}(r_i)\Big|_{\mathcal{X}_G}, \qquad E_{\mathrm{ref},G}(r_i + \delta x) = E_{\mathrm{ref},G}(r_i) + \tfrac{1}{2}\delta x^\top H_i \delta x + o(\|\delta x\|^2). \tag{2}$$

For a unit Cartesian direction e orthogonal to rigid rotations, the directional curvature label is

$$\kappa_i(e) = e^\top H_i e, \qquad \|e\| = 1, \tag{3}$$

with units energy/length^2. This defines stiffness operationally rather than assuming a universal bond–angle–torsion hierarchy. Mode mixing and cross-coordinate coupling are retained. Raw force constants expressed in different internal-coordinate units cannot be ranked without a stated displacement convention.

Ideally $r_i = x_{1,i}$, the training target. Otherwise, relax a separate reference, preserve atom correspondence, and align it to $x_{1,i}$ using a proper rotation. Curvature at r_i is then a transferred local approximation around $x_{1,i}$, not an identity. Record the reference residual force and the aligned displacement $\|r_i - x_{1,i}\|/\sqrt{N}$; do not replace the target silently. Energetic, free-energy, and mass-weighted Hessians have different meanings. The present construction uses an *unweighted Cartesian energy Hessian*; it does not identify its eigenvalues with vibrational frequencies.

2 Rigid-mode separation and quantitative calibration

At r_i, construct rigid rotations from geometry rather than a small-eigenvalue threshold:

$$\mathcal{T}_{\mathrm{rot}}(r_i) = \left\{ (\omega \times r_{i,a})_{a=1}^{N} : \omega \in \mathbb{R}^3 \right\}. \tag{4}$$

Let $Q_{\mathrm{rot},i}$ be an orthonormal basis of this subspace and $Q_{\mathrm{vib},i}$ an orthonormal basis of its complement in $\mathcal{X}_G$. For a non-collinear geometry the rotational rank is three; for a non-degenerate linear geometry it is two. At an exact stationary point of the invariant potential, $H_i Q_{\mathrm{rot},i} = 0$. A nonstationary Hessian need not have exact rotational null modes.

Diagonalize only the vibrational restriction:

$$H_{\mathrm{vib},i} = Q_{\mathrm{vib},i}^{\top} H_i Q_{\mathrm{vib},i} = V_i^{\mathrm{vib}} \,\mathrm{diag}(\lambda_{ij}) (V_i^{\mathrm{vib}})^{\top}. \tag{5}$$

After checking that no materially negative vibrational curvature violates the stable-reference assumption, define

$$\boxed{K_i = Q_{\mathrm{vib},i} f(H_{\mathrm{vib},i}) Q_{\mathrm{vib},i}^{\top} + \kappa_{\mathrm{rot}} Q_{\mathrm{rot},i} Q_{\mathrm{rot},i}^{\top} \succ 0, \qquad f(\lambda) = \max(\lambda, \kappa_{\min}).} \tag{6}$$

Here $\kappa_{\mathrm{rot}}, \kappa_{\min} > 0$ are finite constants in stiffness units. Small numerical negative modes may be regularized under a declared tolerance; a significant unstable mode requires a different reference or an explicitly qualified surrogate. The subspace construction avoids assigning ambiguous eigenvectors to rotations when exact or near-degenerate soft modes are present.

Only unmodified positive vibrational eigenvalues retain their reference calibration. Rotational and floored-mode scales are computational choices. Trace normalization, upper clipping, or condition-number compression would further change that calibration and are not part of this core. Introduce a fixed energy scale and a covariance

$$\varepsilon_E > 0, \qquad C_i^{\mathrm{stiff}} = \varepsilon_E K_i^{-1}. \tag{7}$$

This covariance has squared-length units. Choosing $\varepsilon_E = k_B T$ reproduces the classical *local harmonic* scale on calibrated vibrational modes. It neither assigns basin populations nor makes a minima-trained generator a Boltzmann sampler.

A local consistency check compares an estimated quadratic form to reference energy increments:

$$\mathcal{R}_i(\delta x) = \frac{2\left[E_{\mathrm{ref},G}(r_i + \delta x) - E_{\mathrm{ref},G}(r_i) - \nabla E_{\mathrm{ref},G}(r_i)^{\top} \delta x\right]}{\delta x^{\top} K_i \delta x}. \tag{8}$$

For $\delta x = ae$ and $a \to 0$, this tends to $e^{\top} H_i e / (e^{\top} K_i e)$, provided the denominator is nonzero. It tends to one only in directions whose curvature is accurately retained. This is not a calibration claim for artificial rotational or floored modes. Agreement with an approximate potential is not automatically agreement with a more accurate physical reference.

For a scalar local internal coordinate q at r_i, let $g_i = Dq(r_i)$ be its row Jacobian. A complementary label, including relaxation of coupled degrees of freedom, is

$$\chi_{q,i} = g_i K_i^{-1} g_i^{\top}, \qquad k_{q,i}^{\mathrm{rel}} = \chi_{q,i}^{-1}, \qquad \min_{g_i \delta x = \Delta q} \tfrac{1}{2} \delta x^{\top} K_i \delta x = \frac{(\Delta q)^2}{2\chi_{q,i}}. \tag{9}$$

For $g_i \neq 0$, $\chi_{q,i} > 0$. This is a relaxed quadratic stiffness, not a diagonal Cartesian force constant. Its units are energy/q^2, and its physical interpretation inherits the regularization and reference approximations above. A rotation-invariant q satisfies $g_i Q_{\mathrm{rot},i} = 0$ at the same reference. The minimizing displacement is $\delta x = K_i^{-1} g_i^{\top} \Delta q / \chi_{q,i}$, which proves (9) directly.

3 A common prior and a rotation-compatible target

Choose a graph-dependent covariance satisfying

$$C_{0,G} \succ 0, \qquad \mathcal{R} C_{0,G} \mathcal{R}^\top = C_{0,G}, \qquad S_G = C_{0,G}^{1/2}, \qquad p_0(\cdot \mid G) = \mathcal{N}_{\mathcal{X}_G}(0, C_{0,G}), \tag{10}$$

where the symmetric positive square root fixes a rotation-compatible implementation. An explicit graph-spring example is

$$C_{0,G} = \sigma_0^2 \left[(L_G + \rho I_N)\big|_{\mathbf{1}^\perp} \right]^{-1} \otimes I_3, \qquad \rho > 0. \tag{11}$$

Here L_G and ρ are dimensionless and σ_0 has length units. This source encodes connectivity, not quantitatively calibrated energetic curvature. Structured harmonic sources are relevant molecular precedents [3]; (11) is our explicit choice, not a formula attributed to that work. The ridge handles disconnected graphs algebraically without asserting their physical suitability for a conformer task.

Let the raw joint empirical law of conformers and their stiffness labels be

$$\nu_{\mathrm{emp},G} = \sum_i w_i \, \delta_{(x_{1,i}, K_i)}, \qquad w_i \geq 0, \qquad \sum_i w_i = 1. \tag{12}$$

The weights are the chosen data weights, not weights inferred from $\det K_i$. Define its proper-rotation symmetrization by

$$\boxed{\nu_G = \int_{\mathrm{SO}(3)} (T_R)_\# \nu_{\mathrm{emp},G} \, d\mu(R), \qquad T_R(x_1, K) = (\mathcal{R} x_1, \mathcal{R} K \mathcal{R}^\top),} \tag{13}$$

where μ is normalized Haar measure. In this note $q_{\mathrm{data}}(\cdot \mid G)$ denotes the *conformer marginal of* ν_G. This definition preserves internal conformations and their weights while removing arbitrary stored orientations. It is essential: an equivariant deterministic flow from an invariant source cannot exactly generate a non-invariant set of fixed-orientation coordinates.

Training samples are drawn as

$$G \sim p_{\mathrm{train}}, \qquad (x_1, K) \sim \nu_G, \qquad z \sim \mathcal{N}(0, I_d), \qquad t \sim \mathcal{U}(0,1), \tag{14}$$

with z and t independent of the joint endpoint label. Equivalently, draw an empirical pair and a uniform proper rotation, rotate the pair together, and then draw independent noise. The Hessian and target are *not* sampled independently. Every endpoint has exactly the same source. Atom relabelings must likewise be applied consistently to the graph, geometry, and matrix indices.

4 Stiffness-dependent conditional probability paths

4.1 General family and the revised schedule

Let α, s, h be continuously differentiable schedules satisfying

$$\begin{gathered} \alpha(0) = 0, \quad \alpha(1) = 1, \quad s(0) = 1, \quad s(t) > 0 \ (t < 1), \quad s(1) = 0, \\ h(0) = 0, \quad h(1) = 1, \quad 0 \leq h(t) \leq 1. \end{gathered} \tag{15}$$

For the stronger local endpoint interpretation below, also require $1 - \alpha(t) = o(s(t))$. A fully specified candidate is

$$\boxed{\alpha(t) = 3t^2 - 2t^3, \qquad s(t) = 1 - t, \qquad h(t) = 3t^2 - 2t^3.} \tag{16}$$

Although α and h coincide here, they control different objects: mean progress and precision interpolation. Their equality is a simple choice, not a theoretical necessity. Relative to v1, only the mean schedule changes; the precision and collapse schedules are retained. Section A quantifies the additional mean-path action.

Define

$$\boxed{\begin{aligned} P_{t,K} &= (1-h(t))C_{0,G}^{-1} + h(t)K/\varepsilon_E, \\ m_t &= \alpha(t)x_1, \qquad \Sigma_{t,K} = s(t)^2 P_{t,K}^{-1}, \\ p_t(\cdot \mid x_1, K, G) &= \mathcal{N}_{\mathcal{X}_G}(m_t, \Sigma_{t,K}), \qquad 0 < t < 1. \end{aligned}} \tag{17}$$

Both precision terms have inverse-squared-length units. Positive definiteness holds for every interior time. Thus no finite-curvature direction is frozen. If $C_0 = \sigma_0^2 I$, a stiffness eigenmode κ_j has variance

$$\sigma_{t,j}^2 = \frac{s(t)^2}{(1-h(t))/\sigma_0^2 + h(t)\kappa_j/\varepsilon_E}. \tag{18}$$

At a fixed $t > 0$, larger stiffness gives smaller variance in this commuting example. More generally, $K_a \succeq K_b$ implies $\Sigma_{t,K_a} \preceq \Sigma_{t,K_b}$ for a fixed common source and schedule. This is covariance shaping, not a claim that the whole path follows a low-energy molecular trajectory.

4.2 Why interpolate precision?

For the normalized residual $y = (x - m_t)/s(t)$, let $g_0 = \mathcal{N}(0, C_{0,G})$ and $g_K = \mathcal{N}(0, \varepsilon_E K^{-1})$. The path has residual density

$$r_{t,K}(y) = \frac{g_0(y)^{1-h(t)} g_K(y)^{h(t)}}{Z_{t,K}} = \mathcal{N}(0, P_{t,K}^{-1}). \tag{19}$$

It is the unique density minimizing

$$(1-h)\,\mathrm{KL}(r\|g_0) + h\,\mathrm{KL}(r\|g_K). \tag{20}$$

Indeed, (20) equals $\mathrm{KL}(r\|r_{t,K}) - \log Z_{t,K}$. Thus the precision bridge interpolates two quadratic penalties, or two log densities, in residual space. This variational characterization explains the construction; it does not establish optimal transport, optimal learning, or optimal molecular sampling.

4.3 Shared source and complete endpoint-error calibration

Every conditional starts at p_0, and

$$p_t(\cdot \mid x_1, K, G) \Rightarrow \delta_{x_1}, \qquad \frac{\Sigma_{t,K}}{s(t)^2} \longrightarrow \varepsilon_E K^{-1}. \tag{21}$$

More strongly, for x_t drawn from (17),

$$\frac{x_t - x_1}{s(t)} \;\Big|\; (x_1, K, G) \sim \mathcal{N}\!\left(-\frac{1-\alpha(t)}{s(t)} x_1,\; P_{t,K}^{-1}\right) \;\Rightarrow\; \mathcal{N}(0, \varepsilon_E K^{-1}). \tag{22}$$

The extra condition $1 - \alpha = o(s)$ removes the leading-order mean bias. In (16), $1 - \alpha = (1-t)^2(1+2t)$, so the mean bias is $O(s^2)$ while random displacements are $O(s)$. In contrast, v1 used $\alpha = t$ and $s = 1 - t$, giving the limit $\mathcal{N}(-x_1, \varepsilon_E K^{-1})$ for the normalized *full* error. Its conditional covariance was correct, but not a zero-mean complete local perturbation.

This improvement is asymptotic. At intermediate times $m_t = \alpha(t)x_1$ still uniformly scales the molecule, and the Gaussian has broad support outside a reference basin. Neither the revised mean nor a local Hessian makes the entire path physical.

5 Conditional velocity without a commutation assumption

Whiten the stiffness using the common source square root:

$$B_K = S_G^\top K S_G/\varepsilon_E, \qquad D_{t,K} = (1-h)I + hB_K, \qquad A_{t,K} = sS_G D_{t,K}^{-1/2}. \tag{23}$$

Then

$$P_{t,K} = S_G^{-\top} D_{t,K} S_G^{-1}, \qquad A_{t,K}A_{t,K}^\top = s^2 S_G D_{t,K}^{-1} S_G^\top = \Sigma_{t,K}. \tag{24}$$

For a time-independent sample $z \sim \mathcal{N}(0, I_d)$, set

$$\boxed{x_t = \alpha(t)x_1 + A_{t,K}z, \qquad U_t = \dot\alpha(t)x_1 + \dot A_{t,K}z.} \tag{25}$$

Diagonalize $B_K = V\,\mathrm{diag}(b_j)V^\top$ and write $d_j(t) = (1-h(t)) + h(t)b_j$. Since D and $\dot D$ are functions of the same fixed B_K,

$$\dot A_{t,K} = S_G V\,\mathrm{diag}\Big[\dot s\, d_j^{-1/2} - \tfrac{1}{2}s\dot h\,(b_j-1)d_j^{-3/2}\Big]\,V^\top. \tag{26}$$

This derivative does not assume $C_{0,G}K = KC_{0,G}$. For the default schedules, $\dot s = -1$ and $\dot\alpha = \dot h = 6t(1-t)$. For $0 < t < 1$, A is invertible and the corresponding Eulerian conditional field is

$$\boxed{\begin{aligned} u_t(x \mid x_1, K, G) &= \dot\alpha(t)x_1 + M_{t,K}(x - \alpha(t)x_1), \\ M_{t,K} = \dot A A^{-1} &= \frac{\dot s}{s}I - \frac{1}{2}P_{t,K}^{-1}\dot P_{t,K}, \\ \dot P_{t,K} &= \dot h\left(K/\varepsilon_E - C_{0,G}^{-1}\right). \end{aligned}} \tag{27}$$

The order $P^{-1}\dot P$ is essential. It is generally not interchangeable with $\dot P P^{-1}$. Equivalently, $M = (\dot s/s)I - (\dot h/2)S_G D^{-1}(B_K - I)S_G^{-1}$.

Proposition 1 (Conditional continuity)**.** *For each fixed endpoint label, the field in* (27) *transports the Gaussian path* (17) *on* $0 < t < 1$.

Proof. The field gives $\dot m_t = \dot\alpha x_1$. Moreover,

$$\dot\Sigma = 2\frac{\dot s}{s}\Sigma - s^2 P^{-1}\dot P P^{-1} = M\Sigma + \Sigma M^\top. \tag{28}$$

These are precisely the mean and covariance equations of an affine transport of a Gaussian; alternatively, they follow by differentiating the explicit sample map in (25). Hence $\partial_t p_t + \nabla_{\mathcal{X}_G}\cdot(p_t u_t) = 0$. □

The factor A selects one valid coupling through the covariance path; the covariance alone does not specify a unique velocity. Train using the sample target U_t, not the expression containing $1/s$. For the default schedule and a finite positive spectrum,

$$\lim_{t\to 1}\dot A_{t,K} = -S_G B_K^{-1/2}, \qquad \lim_{t\to 1} U_t = -S_G B_K^{-1/2} z. \tag{29}$$

A finite sampled target does not remove the singular Eulerian conditional gradient as the density collapses, nor bound its size uniformly over arbitrarily extreme spectra.

6 Marginal flow matching, symmetry, and inference

The network observes only (x_t, t, G), not the endpoint or its stiffness label. With the joint sampling law (14), minimize the ordinary Euclidean loss

$$\boxed{\mathcal{L}_{\text{StifFlow}}(\theta) = \mathbb{E}\left\| v_\theta(x_t, t; G) - \left(\dot{\alpha}(t)x_1 + \dot{A}_{t,K}z\right)\right\|^2.} \tag{30}$$

Assume finite target second moments and sufficient integrability to interchange differentiation and averaging on compact time intervals inside $[0, 1)$. These hold for a finite set of finite, positive-definite labels and their rotation orbits under the stated schedules.

Proposition 2 (Marginal transport)**.** *The unrestricted population minimizer and its density path are*

$$\begin{aligned} v^*(x, t; G) &= \mathbb{E}[U_t \mid x_t = x, t, G], \\ p_t(x \mid G) &= \int p_t(x \mid x_1, K, G)\, \nu_G(dx_1, dK), \\ \partial_t p_t + \nabla_{\mathcal{X}_G} \cdot (p_t v^*) &= 0. \end{aligned} \tag{31}$$

The path starts at p_0 and converges to q_{data} as $t \to 1$.

Proof. Squared-error regression gives the conditional expectation. Given (x_1, K) and an interior x_t, the invertibility of A identifies z and therefore $U_t = u_t(x_t \mid x_1, K, G)$. Consequently

$$p_t(x \mid G)v^*(x, t; G) = \int p_t(x \mid x_1, K, G)u_t(x \mid x_1, K, G)\, d\nu_G.$$

Integrating the conditional continuity equations proves (31), the standard CFM marginalization argument [1]. The common initial conditional law gives p_0; terminal convergence follows from the explicit endpoint coupling. In particular, whenever the following moments are finite,

$$W_2^2(p_t(\cdot \mid G), q_{\text{data}}(\cdot \mid G)) \le (1 - \alpha(t))^2\mathbb{E}_{\nu_G}\|x_1\|^2 + s(t)^2\mathbb{E}_{\nu_G}\,\mathrm{tr}(P_{t,K}^{-1}) \longrightarrow 0. \tag{32}$$

Here W_2 is the quadratic Wasserstein distance. The convergence holds for the stated finite empirical law; more general label laws require uniform integrability of the coupled errors. The inequality uses a specific coupling, not an optimality claim for the conditional path. □

Symmetry. Use a centered SO(3)-equivariant velocity network. By source invariance, its symmetric root commutes with $\mathcal{R}$, so A and M transform by conjugation when K is rotated. Thus

$$u_t(\mathcal{R}x \mid \mathcal{R}x_1, \mathcal{R}K\mathcal{R}^\top, G) = \mathcal{R}u_t(x \mid x_1, K, G), \tag{33}$$

and the symmetrized law (13) implies $v^*(\mathcal{R}x, t; G) = \mathcal{R}v^*(x, t; G)$. No reflection symmetry is imposed for a specified chiral graph. This establishes compatibility of the ideal regression target with the architectural symmetry, not chemical validity or guaranteed chirality preservation.

Inference. Sample and integrate

$$z_0 \sim \mathcal{N}(0, I_d), \qquad x(0) = S_G z_0, \qquad \frac{dx}{dt} = v_\theta(x, t; G). \tag{34}$$

No target Hessian or endpoint is needed at inference. Under standard ODE well-posedness conditions on every interval $[0, T]$, $T < 1$, the ideal field generates p_T. Neural optimization, finite capacity, and numerical integration introduce separate errors. The rotation-symmetrized finite empirical target is

generally singular in $\mathcal{X}_G$; exact transport to it is a terminal limit, not the endpoint of a globally smooth invertible flow with bounded derivatives. Stopping at $1-\eta$ or retaining nonzero terminal noise changes the ideal target and should be stated explicitly.

The derivation assumes independent source noise and endpoint labels. An endpoint-dependent alignment or a different source–target coupling must be re-derived rather than inserted into the Gaussian formulas unchanged. Reference–target alignment used to transfer K is a different operation and does not align the sampled source.

7 Local flexibility and the actual learning hypothesis

In this section all limits condition on $(x_{1,i}, K_i, G)$. Let $q(x)$ be a local internal-coordinate map and $J_i = Dq(x_{1,i})$. Use a smooth local chart for angular coordinates, away from degenerate geometries. From (22) and first-order Taylor expansion,

$$\frac{q(x_t) - q(x_{1,i})}{s(t)} \;\Rightarrow\; \mathcal{N}\Big(0, \varepsilon_E J_i K_i^{-1} J_i^\top\Big). \tag{35}$$

If the Taylor remainder is negligible in mean square—for example, for a smooth extension of the local chart with suitable growth control—then also

$$\mathrm{Cov}(q(x_t) \mid x_{1,i}, K_i, G) = s(t)^2 \varepsilon_E J_i K_i^{-1} J_i^\top + o(s(t)^2) = J_i \Sigma_{t,K_i} J_i^\top + o(s(t)^2). \tag{36}$$

For a scalar coordinate with a nonzero derivative, $J_i K_i^{-1} J_i^\top > 0$. Bond stretching and angle bending therefore retain positive *leading local* variance; coupled motion is allowed by off-diagonal terms. A redundant vector of internal coordinates can have a singular covariance even when each nonconstant scalar component fluctuates. If $r_i = x_{1,i}$ and q is rotation-invariant, $J_i Q_{\mathrm{rot},i} = 0$, so the artificial rotational covariance drops out to first order. When reference curvature is transferred from $r_i \neq x_{1,i}$, that cancellation need not be exact.

These limits say nothing about global torsional barriers or final thermal variance. The conditional noise vanishes at the endpoint, and the final conformer populations and fluctuations are supplied by q_{data}, not generated from K_i or ε_E.

Importantly, *any* admissible positive-definite label collection with the same conformer marginal gives the same ideal endpoint distribution. Quantitative stiffness can therefore matter through finite-data, finite-capacity, or finite-computation behavior, not through correctness of the endpoint theorem alone. The precise regression decomposition is

$$\mathcal{L}_{\mathrm{StifFlow}}(\theta) = \mathbb{E}\|v_\theta(x_t, t; G) - v^*(x_t, t; G)\|^2 + \mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}(U_t \mid x_t, t, G). \tag{37}$$

Changing the path changes both the target field and this irreducible conditional variance. A lower raw CFM loss across different paths is consequently not, by itself, proof of a better fitted transporting velocity. The research question is whether *quantitatively calibrated* conditional geometry improves approximation of the marginal conformer-generating field enough to offset the added path complexity. The mathematical construction does not settle that question.

8 Scope and relation to existing formulations

The core consists of a specified energetic stiffness label, an invariant common prior, an endpoint-conditioned precision bridge, an analytic conditional velocity, and ordinary CFM. Hessian-informed flow

matching already incorporates local energy curvature into conditional flows using linearized dynamics and time transformations [2]. ET-Flow provides a molecular precedent for equivariant Cartesian flow matching with a harmonic prior [3]. Riemannian denoising for molecular optimization provides a related example of geometry informed by energetic changes [4]. None of these citations is a novelty certificate for this construction.

StifFlow here is a *stiffness-informed conditional-path design*, not a physical molecular dynamics model, a Riemannian geodesic construction, or a proven numerical preconditioner. Its precise guarantees are the common source, positive interior covariance, conditional and marginal continuity, the rotation symmetry, and the local terminal limits under the stated assumptions. Energy calibration outside the reference basin, molecular validity, generalization, coverage, and reduced numbers of function evaluations (NFE) are not guaranteed. The following appendices record concrete costs, excluded extensions, and reproducible algebra checks.

A Schedule costs, locality, and mathematical baselines

A.1 A local endpoint schedule is not a free simplification

For a fixed pair (x_1, K), define the conditional Euclidean squared-speed integral

$$\mathcal{A}(x_1, K) = \int_0^1 \mathbb{E}_z \|\dot{x}_t\|^2 \, dt = \|x_1\|^2 \int_0^1 \dot{\alpha}(t)^2 dt + \int_0^1 \|\dot{A}_{t,K}\|_F^2 dt. \tag{38}$$

This is an artificial-time path functional, not physical kinetic energy. Using $\alpha(1) - \alpha(0) = 1$ and $A_1 - A_0 = -S_G$ gives the exact identity

$$\boxed{\begin{aligned}\mathcal{A}(x_1, K) = \quad & \underbrace{\|x_1\|^2 + \|S_G\|_F^2}_{\text{straight path, same endpoint pairing}} \\ & + \|x_1\|^2 \int_0^1 (\dot{\alpha} - 1)^2 dt + \int_0^1 \|\dot{A} + S_G\|_F^2 dt.\end{aligned}} \tag{39}$$

Both excess terms are nonnegative. For the revised smoothstep mean,

$$\int_0^1 \dot{\alpha}^2 dt = \frac{6}{5}, \qquad \int_0^1 (\dot{\alpha} - 1)^2 dt = \frac{1}{5}. \tag{40}$$

Thus the mean contribution is 20% higher than for $\alpha = t$; covariance deformation can add further cost. This is the explicit trade-off for removing the leading endpoint mean bias. It does not imply that the marginal flow is harder to learn: Jensen's inequality only gives $\int \mathbb{E}_{p_t} \|v^*\|^2 dt \le \mathbb{E}_{\nu_G} \mathcal{A}$, not an ordering of the marginal actions for two different designs.

A.2 Large whitened stiffness can act at early times

The dimensionless eigenvalues b_j of B_K compare energetic stiffness with the source scale. In the scalar case $C_0 = \sigma_0^2$ and $b = \sigma_0^2 \kappa / \varepsilon_E$,

$$a_b(t) = \frac{\sigma_0 (1 - t)}{\sqrt{1 + h(t)(b - 1)}}. \tag{41}$$

For $b \gg 1$ and the default h, the precision correction is already order one at $t = O(b^{-1/2})$, despite $\dot{h}(0) = 0$. Substituting $t = c/\sqrt{b}$ gives

$$\dot{a}_b(t) = -\sigma_0 \frac{3c\sqrt{b}}{(1 + 3c^2)^{3/2}} + O(\sigma_0), \qquad \max_t |\dot{a}_b(t)| \sim \frac{2}{3} \sigma_0 \sqrt{b}. \tag{42}$$

The maximum of the leading coefficient occurs at $c = 1/\sqrt{6}$. Early large targets and broad perturbations can therefore persist even with the revised mean. For non-isotropic sources, the Euclidean behavior also depends on S_G, not only on the eigenvalues of B_K. Keeping physical eigenvalues does not automatically reduce numerical stiffness.

A delayed schedule is an *optional*, explicitly different design: for fixed $t_{\text{on}} \in [0, 1)$,

$$h_{\text{delay}}(t) = \begin{cases} 0, & t \le t_{\text{on}}, \\ 3u^2 - 2u^3, \quad u = (t - t_{\text{on}})/(1 - t_{\text{on}}), & t > t_{\text{on}}. \end{cases} \tag{43}$$

All main derivations still hold. Delaying onset can prevent the stiffness label from acting before a chosen time, but compresses its change into a shorter interval. It neither proves locality of the Gaussian samples nor removes potentially large velocity targets. It is not part of the fixed default in (16).

A.3 Baseline identities and computational meaning

Setting $h \equiv 0$ makes $A_t = s(t)S_G$ and removes stiffness dependence. With the revised α this is a *stiffness-free, schedule-matched* path, not a straight interpolation. The ordinary straight path is recovered only when also setting $\alpha = t$ and $s = 1 - t$. Thus removing stiffness and changing the mean schedule are distinct mathematical operations.

In a dense implementation, curvature estimation is a separate reference-model cost. Storing K or its factorization costs $O(d^2)$ per label and dense diagonalization costs $O(d^3)$. Once B_K is diagonalized, changing time only changes d scalar coefficients; dense matrix–vector applications remain $O(d^2)$ per sampled label. These are operation-count statements, not timings. The absence of an inference-time Hessian does not make preprocessing or training-label evaluation free.

B Optional extensions that are not the core

A seed-mixture prior. If an inference-time seed distribution $\pi(dr \mid G)$ and curvature estimates at sampled seeds are available, an alternative source is

$$p_0^{\text{seed}}(x \mid G) = \int \mathcal{N}_{\mathcal{X}_G}(x; r, \varepsilon_E K(r)^{-1})\, \pi(dr \mid G). \tag{44}$$

Its marginal density need not be tractable. Source symmetry must be enforced consistently, and a narrow seed law may limit exploration. This is a different prior with additional costs, not a requirement of StifFlow's common graph-Gaussian core.

A loss metric. For a fixed, θ-independent SPD weight $W = W(x_t, t, G)$ measurable from the network inputs,

$$\mathcal{L}_W = \mathbb{E}[(v_\theta - U_t)^\top W(x_t, t, G)(v_\theta - U_t)] \tag{45}$$

has the same unrestricted population optimum v^*. It can nevertheless change the best approximation within a finite function class and the optimization behavior. Physical units must be handled explicitly; this is a loss norm, not a construction of Riemannian geodesics.

If the weight instead depends on the hidden endpoint label, the optimum is

$$v_W^*(x, t; G) = \big(\mathbb{E}[W_i \mid x, t, G]\big)^{-1} \mathbb{E}[W_i U_t \mid x, t, G], \tag{46}$$

which generally differs from the required conditional expectation. A two-component example with equal posterior probabilities, scalar targets -2 and 2, and weights 1 and 3 gives $v^* = 0$ but $v_W^* = 1$. Thus

hidden-endpoint Hessian weighting is not interchangeable with ordinary CFM. Multiplying an already learned velocity by an inverse-stiffness matrix also generally changes its continuity equation and endpoint law; it is not a free numerical preconditioner.

C Reproducible algebra checks and revision record

The companion file `StifFlow_core_formulation_v2_checks.py` independently checks the *revised* schedules. It uses NumPy and SciPy, random seed 20260915, 12 noncommuting SPD matrix pairs in dimensions 3, 5, and 9, and 60 interior time points. Derivative checks use a fourth-order centered difference with step 10^{-5}. The accompanying JSON contains full parameters, normalization conventions, scalar examples, and all recorded errors.

Check	Maximum error
Covariance factor $AA^\top = \Sigma$ (normalized)	3.2×10^{-15}
Sample derivative against $\dot\alpha x_1 + \dot A z$ (absolute)	8.7×10^{-11}
Compact velocity matrix M (normalized)	2.0×10^{-15}
Covariance evolution (normalized)	1.3×10^{-11}
Gaussian continuity residual (normalized)	1.4×10^{-11}
Molecular SO(3) path covariance (normalized)	1.4×10^{-15}
Full conditional action identity (normalized)	2.5×10^{-16}
KL interpolation identity (absolute)	2.3×10^{-16}

The script also checks the common source, complete normalized endpoint second moment, relaxed-coordinate quadratic minimization, a local bond-length limit, and geometric rigid-mode separation for stationary invariant synthetic spring potentials. Scalar diagnostics with $\sigma_0 = 1$ give $\max_t |\dot a_b| \approx 7.03$ for $b = 100$ and 67.03 for $b = 10^4$, consistent with (42). These are synthetic algebra checks, not molecular calibration data, neural training results, or evidence of better generation.

Revision record. The main schedule change from v1 is $m_t = tx_1 \mapsto \alpha(t)x_1$, with supervision $\dot\alpha x_1 + \dot A z$. The common source and precision bridge are retained. The joint target law, rigid-mode projector, complete endpoint-error limit, local-coordinate assumptions, and action costs are now explicit. New variational and schedule results are derived here, not attributed to the cited papers.

References

[1] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. *Flow Matching for Generative Modeling.* International Conference on Learning Representations (ICLR), 2023. arXiv:2210.02747; mathematical reference: version 2.

[2] C. I. Sprague, A. Elofsson, and H. Azizpour. *Hessian-Informed Flow Matching.* arXiv:2410.11433v1, 2024.

[3] M. Hassan, N. Shenoy, J. Lee, H. Stärk, S. Thaler, and D. Beaini. *ET-Flow: Equivariant Flow-Matching for Molecular Conformer Generation.* Advances in Neural Information Processing Systems (NeurIPS), 2024. arXiv:2410.22388.

[4] J. Woo, S. Kim, J. H. Kim, and W. Y. Kim. *Riemannian Denoising Model for Molecular Structure Optimization with Chemical Accuracy.* Nature Computational Science **6**, 134–144, 2026. doi:10.1038/s43588-025-00919-1.